

4 The gene for the enzyme catalase is on chromosome 11 in humans.

(a) Explain the meaning of the term gene.

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..... [2]

(b) Two enzymes, DNA polymerase and DNA ligase, are involved in the replication of DNA.

Fig. 4.1 shows the replication of part of human chromosome 11 by DNA polymerase. The arrows show the direction of synthesis of the new polynucleotides by DNA polymerase.

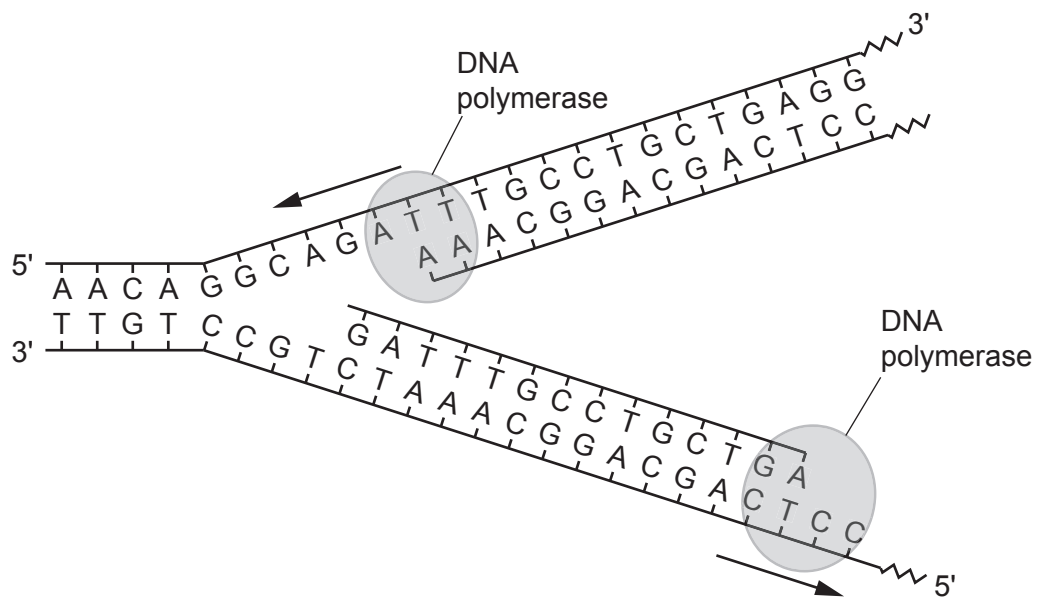


Fig. 4.1

(i) Describe the roles of DNA polymerase and DNA ligase in the replication of DNA.

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..... [5]

(ii) State the name of the stage of interphase in the cell cycle when DNA replication occurs.

..... [1]

(c) Fig. 4.2 is a diagram of chromosome 11 at metaphase of mitosis.

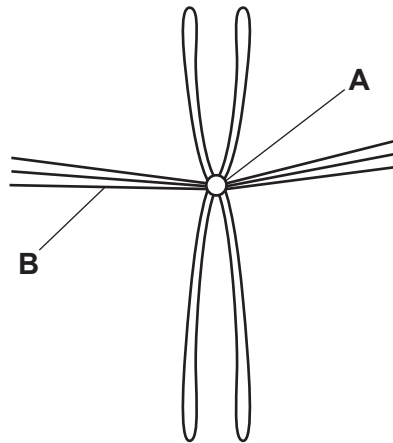


Fig. 4.2

(i) State the names **and** functions of structures **A** and **B**.

structure **A**

function

.....

structure **B**

function

.....

[2]

(ii) Complete Fig. 4.3 to show what happens to chromosome 11 in anaphase, so that the daughter nuclei are genetically identical.

cell surface
membrane

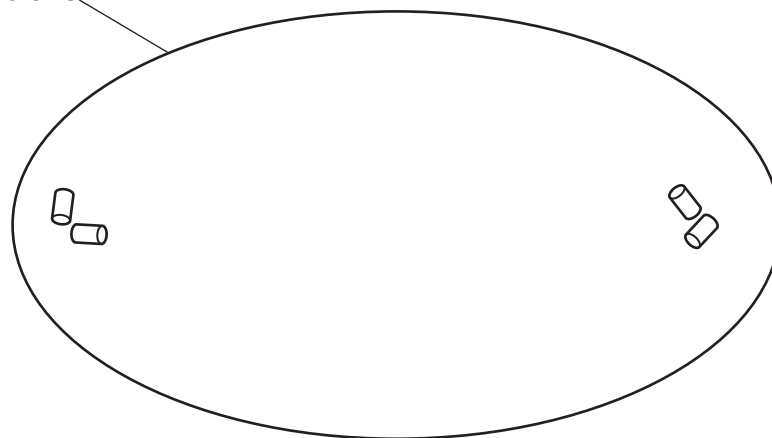


Fig. 4.3

[3]

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Question 5 is on page 16.